

A LECTURE ON

How to Read Research Papers.

The math, the triage, the bullshit detection.

PART 1

Why Papers, Why Now

The field moves in weeks.

Why papers, why now.

- The field moves in weeks, not years.
- Tweets are downstream of papers.
- The 10% rule.

The 10% rule.

- 90% of papers don't matter.
- 10% change how you build.
- The skill is identifying the 10% fast.

What you're optimizing for.

- Not "read every paper."
 - Triage > comprehension.
 - Intuition > rigor.
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PART 2

Anatomy of an ML Paper

Where everything lives — and hides.

The standard ML paper.

- 01 Abstract
 - 02 Introduction
 - 03 Related work
 - 04 Method
 - 05 Experiments / Results
 - 06 Ablations
 - 07 Conclusion
 - 08 Appendix
-

Abstract.

- The whole paper in 200 words.
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- Claims + the headline number.
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- Read it first, always.

Introduction.

— Motivation, and the gap.

— "Here's what we did."

— Contribution bullets.

Related work.

- Author signaling.
- Usually skippable.
- Read only if you don't know the field.

Method.

- The actual technique.
- The math lives here.
- Read it after the results.

Experiments.

- Setup, datasets, baselines.
- Where setups get vague.
- Watch for what's missing.

Results & ablations.

- Headline tables and figures.
-
- Ablations are the honest part.
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- Cherry-picking lives here too.

Conclusion & appendix.

- Conclusion: throwaway.
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- Appendix: where the real details hide.
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- Hyperparameters, failure cases, extra ablations.

Where authors hide what.

Bad results

buried in the appendix

Cherry-picked examples

the main figures

Missing baselines

silently absent

Caveats

the footnotes

PART 3

The Triage Pass

Linear reading is for textbooks.

Read non-linearly.

- Linear reading is for textbooks.
- Papers are reference docs.
- Jump to the signal.

The order.

- 01 Abstract
- 02 Figures + captions
- 03 Results table
- 04 Method overview (skim)
- 05 Ablations
- 06 Then decide if you go deeper

Three-pass method (Keshav).

Pass 1

skim — 10 minutes

Pass 2

details — 1 hour

Pass 3

reimplement — 4+ hours

Reality

most papers stop at pass 1

The decision.

- Does it change how I build?
- Is the method novel, or repackaged?
- Are the claims verifiable?
- If no — close it.

PART 4

Red Flags & What to Skip

Skim, skip, study.

Red flags.

- Single benchmark.
- Missing baselines.
- No code.
- Cherry-picked examples.
- Vague setup.
- "We achieve SOTA."

Single-benchmark wins.

- One leaderboard $\neq$ general capability.
-
- Ask: does it work on 3+ unrelated tasks?

Missing baselines.

- No comparison to the obvious priors.
- "We outperform Llama-3" — at what size? what budget?
- Apples-to-apples, or it doesn't count.

No code.

- Cannot be verified.
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- Treat the claims as conjectures.
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- Especially in industry papers.

Cherry-picked examples.

- Main-figure prompts $\neq$ random samples.
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- Look for "we report random samples in the appendix."
-
- If absent — assume curation.

Vague setup.

- Missing hyperparameters.
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- "Trained on a large corpus" — which one?
-
- Reproducibility is an honesty signal.

"We achieve SOTA."

- SOTA is a marketing word.
 - Ablations matter more than the headline.
 - Ask: what's the actual delta?
-

What to skip without guilt.

- Related work (usually).
- Most proofs.
- Hyperparameter appendices.
- Author bios.
- Acknowledgements.

Skim / skip / study.

Skim	<i>intro, method overview</i>
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Skip	<i>related work, most proofs</i>
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Study	<i>results, ablations, the key formula</i>
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PART 5

Math Survival Kit:

Notation

Pattern-match before parsing.

The ten notations.

$\mathbb{E}$	<i>Expectations</i>
D_{KL}	<i>KL divergence</i>
∇	<i>Gradients</i>
$\text{argmax} / \text{argmin}$	<i>the input that maximizes / minimizes</i>
π / π_{old}	<i>Ratios</i>
$p(y \mid x)$	<i>Conditionals</i>
$\Sigma \quad \Pi$	<i>Sums and products</i>
$\mathbb{1}$	<i>Indicators</i>
$O(\cdot)$	<i>Big-O</i>
$\mathcal{L}$	<i>Lagrangians</i>

Expectations & KL.

$\mathbb{E}[X]$

"the average of X over the distribution"

$D_{\text{KL}}(P \parallel Q)$

"how different are P and Q "

Gradients & argmax.

∇L

"direction of steepest increase of L "

$\operatorname{argmax}_x f(x)$

"the x that maximizes f "

optimization

is just chasing these

Ratios & conditionals.

$\pi(a|s) / \pi_{old}(a|s)$

"how much has the policy changed"

$p(y | x)$

"distribution of y given we saw x"

language modeling

is all $p(\text{token} | \text{context})$

Sums & indicators.

 Σ_i *"loop"* $\prod_i$ *"product loop"* $\mathbb{1}[\text{cond}]$ *1 if true, 0 otherwise***all of them***for-loops in disguise*

Big-O & Lagrangians.

$O(n^2)$

"scales quadratically with n "

$\mathcal{L} = f + \lambda \cdot g$

"optimize f subject to g "

λ

the price of the constraint

Form matters — pattern-match first.

$$\theta_{t+1} = \theta_t - \alpha \cdot (\cdot)$$

an optimization step

$$L = -\log p(y|x)$$

cross-entropy loss

$$\pi / \pi_{\text{old}}$$

policy ratio (PPO family)

PART 6

The 5-Step Method for Math

30 minutes per formula. Worth it.

The five-step method.

- Our framework for any formula.
 - Use it on every math-heavy paper.
 - About 30 minutes per formula. Worth it.
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Step 0 — Breathe.

- Overwhelm is normal.
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- You're not supposed to glance and get it.
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- Seniors feel this too.

Step 1 — Identify.

- All formulas in the paper.
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- Including ones referenced but not shown.
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- Group them by logical block.

Step 2 — Move to paper.

— Off the screen.

— Real pen, real paper.

— Degrees of freedom matter.

Step 3 — Translate.

- Read meaning, not symbol.
-
- "Learning rate," not "alpha."
-
- Name every variable in your head.

Step 4 — Edge cases.

- Set the knobs to 0 and 1.
 - What known algorithm do you recover?
 - "Just X with one extra knob" is the insight.
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Step 5 — Distill.

- One-sentence intuition per formula.
- Not rigorous — logical.
- Rubber-duck test: can you explain it?

Primitives → novelty.

- Papers build vocabulary.
 - Section 5 ≠ Section 0, for a reason.
 - Lost in the method? Back up to the primitives.
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PART 7

R₁ + GRPO Deep Dive

One formula. 25 minutes. Live.

You pre-read R1.

- Worksheet in hand.
- Triage findings ready.
- Now we walk through it together.

Triage discussion.

- What did the accuracy curve show?
 - The "aha moment" — real, or marketing?
 - Red flags you spotted?
 - What did you skip?
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R1-Zero vs. R1.

R1-Zero

pure RL on the base model

R1

cold-start SFT $\rightarrow$ RL $\rightarrow$ SFT $\rightarrow$ RL

the point

R1-Zero is the experiment, not the product

The reward design.

- Accuracy reward (a verifier).
 - Format reward (<think>...</think>).
 - No reward model.
 - Why this is the whole story.
-

Cold-start data.

- Why R1-Zero alone wasn't shipped.
- Readability, language mixing.
- A few thousand examples.
- The SFT–RL sandwich.

The GRPO formula.

One formula.

25 minutes.

The five-step method, live.

Steps 1–2 — Identify + paper.

- Pull the GRPO objective off the page.
 - Onto the board, or your notes.
 - List every symbol.
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Step 3 — Name every symbol.

π_θ	<i>current policy</i>
π_{old}	<i>snapshot of the policy before the update</i>
A_i	<i>advantage for sample i</i>
ϵ	<i>clip bound</i>
β	<i>KL coefficient</i>

Step 4 — Edge cases.

$\beta = 0$

no KL anchor

$A_i = \text{reward directly}$

no group baseline

what you recover

PPO

so GRPO is

PPO with group-relative advantage

Step 5 — Distill.

"Sample a group. Advantage = z-score within the group. Clipped policy gradient, KL anchor."

No value model needed. That's it.

PART 8

Building a Habit

Read 1, skim 5. Every week.

One paper a week.

Cadence	<i>read 1, skim 5</i>
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Slot	<i>Tuesday morning, hour blocked</i>
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Cumulative	<i>50 a year</i>
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Feeds.

- alphaXiv.
- Hugging Face Daily Papers.
- arXiv-sanity.
- Don't subscribe to everything.

Paper clubs.

- Three to five people.
- One presenter, weekly.
- Forces you to read deeper.
- Best ROI in the field.

X as a filter.

- Trusted accounts curate for you.
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- A good thread is pass-1, done for you.
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- Read the thread first, the paper if interested.

Closing.

- Most papers don't matter.
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- The ones that do deserve the five-step method.
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- This is a skill. Practice is the only path.